International Studies

Georgia Institute of Technology - Group 2 Social / Behavioral Research Investigators and Key Personnel

International Studies

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This module provides information for U.S. researchers and/or collaborating international research investigators who receive funding from the U.S. federal government sources and plan to conduct human subject research outside the U.S.. The module is focused on international research ethics, U.S. government guidelines, and international guidelines. It also includes a listing of ethical review links for countries or regions of the world. It is intended to help researchers and their staff members identify ethical requirements of their global research partners.

Introduction

This module provides information for U.S. researchers and collaborating foreign investigators who receive funding from U.S. government sources and plan to conduct

research outside the U.S. using human subjects. Although the focus of this module is research done in developing countries, these considerations will be applicable to U.S. government funded human subjects research conducted in any country.

Learning Objectives

By the end of this module, you should be able to:

- Describe ethical issues that may affect planning research outside the U.S.
- Identify published international research ethics guidelines.
- Describe specific ethical issues that have been raised in international research.
- Describe U.S. government regulations for ethical review of international projects.
- Understand the responsibilities of researchers seeking ethical review in host countries.

International Research Ethics

Interpreting Core Ethical Concepts



The bioethical precepts of the *Belmont Report* (autonomy, beneficence, and justice) and the four Principles of Bioethics described by Beauchamp and Childress (autonomy, non-maleficence, beneficence, and justice) provide a framework for conducting human subjects research in the U.S. and other Western Countries (Beauchamp and Childress 2001). Ethicists from around the world are debating the "Core" concepts embodied in these documents, comparing and contrasting them to the historical ethics in non-western cultures (for example, Islamic ethics and Confucian ethics) (Aksoy and Elmai 2002, 211-24; Tsai 1999, 315-21; Zhaojian 1995, 239-46). A clear understanding of

relevant cultural issues in host countries will provide a guide for the design and conduct of human subjects research studies in developing countries.

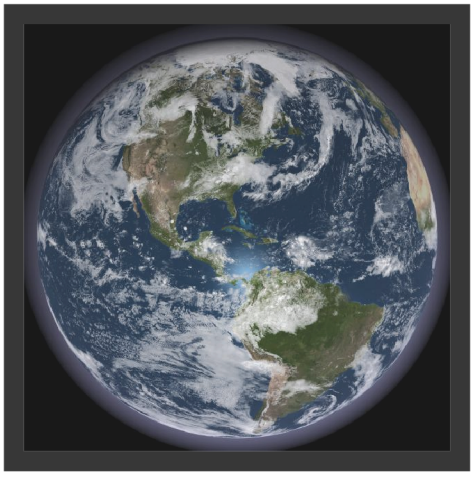
The interpretation of one or more of the core principles and what constitutes ethical behavior may differ from country to country. The ethical principle of respect for persons as described in the *Belmont Report* may not be emphasized to the degree it is in countries that focus on individualism. In many societies, decision-making is a shared responsibility between the family members and the community. For example, in some populations an individual cannot make a decision about medical care without the full engagement of relatives, friends, and/or those in authority in the community. Investigators will need to skillfully design their studies to ensure they are in compliance with U.S. federal and state policies, laws and regulations and at the same time respectful, sensitive, and responsive to local beliefs and practices.

Research Ethics in Developing Countries

Many have questioned the practice of institutions from Western countries sponsoring and/or conducting research with humans in developing countries. The recent debate over establishing international ethical guidelines for human subjects research has been intense and enhanced by the diverse opinions of the stakeholders around the world (see Additional Resources section below). As a consequence of the international debate several consensus documents have been developed.

International Guidelines

- The **Council for International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines for Health-related Research Involving Humans** , originally developed in 1982, has been most recently revised in 2016. These revised guidelines were prepared under the auspices of the World Health Organization (WHO) and reflect the changes, advances, and controversies that have marked biomedical research ethics in the last two decades. Available in English and Spanish.



- The ethics of research related to healthcare in developing countries, the **Nuffield Council on Bioethics** report, was published in April 2002. A follow-up report was published in 2005. The purpose of this report was to examine the ethical issues raised when research related to healthcare is carried out in developing countries and funded by sponsors from more developed countries. View the report and follow-up report.
- **International Guidelines for Ethical Review of Epidemiological Studies**, prepared by CIOMS in cooperation with WHO and the Global Programme on AIDS, was released in 1991. These guidelines are intended for investigators, health policy-makers, members of ethical review committees, and others who need to address ethical issues that arise in epidemiological studies. Access the guidelines.
- Guidelines for biomedical HIV prevention trials have been published at http://www.avac.org/good-participatory-practice.
- In 1995, WHO adopted the **World Health Organization's Good Clinical Practice Guideline (WHO GCP)**. The purpose of the WHO GCP guidelines for trials on pharmaceutical products is to set globally applicable standards for the conduct of such biomedical research on human subjects. They are intended to be applied during all stages of drug development both prior to and after product registration and marketing, but they are also applicable to biomedical research in general. Review the WHO GCP guidelines.
- The **International Council on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH)** brings together the regulatory authorities of Europe, Japan, and the United States and experts from the pharmaceutical industry to discuss scientific and technical aspects of product registration. The purpose is to make recommendations on ways to achieve greater agreement in interpretation and application of technical guidelines and requirements for product registration. The goals are to reduce the need for duplicate testing during research and development of new medicines by more economical use of human, animal and material resources, and to eliminate unnecessary delay in development and availability of new medicines, while

safeguarding quality, safety and efficacy, and compliance with regulatory obligations. Available in English.

- In 2000, WHO developed **Operational Guidelines for Ethics Committees that Review Biomedical Research**, intended to facilitate and support ethical review in all countries around the world. These guidelines are based on a close examination of the requirements for ethical review as established in international guidelines, as well as on an evaluation of existing practices of ethical review in countries around the world. Review these guidelines.
- **Report and Recommendations of the U.S. National Bioethics Advisory Commission, April 2001**. The Commission's attention was focused on the conduct of clinical trials involving competent adults, particularly Phase III drug trials done in other countries by U.S. researchers. In particular, this report focuses on the need for such trials to be directly relevant to the health needs of the host country. However, much of the discussion is relevant to other types of research. Read the report.
- NIAID has launched the ClinRegs website, an online database of country-specific clinical research regulatory information. ClinRegs is designed to allow researchers to plan international clinical trials more efficiently by consolidating regulatory requirements in one place and enabling comparisons across countries.
- OHRP provides listing of international research standards. The International Compilation of Human Research Standards lists over 1,000 laws, regulations, and guidelines from around the world. The Listing of Social and Behavioral Research Standards provides a summary of international research standards specifically for social and behavioral research developed in 18 countries and one international organization.

Protection of Subjects

- Informed consent (Moodley 2010, 197-215; Farnell 2002, 201-2; Porter 1996, 237-43)
- Vulnerable populations (Charo 1993, 135-87; Fitzgerald and Behets 2003, 68-9)
- Research with children (Kauffman 2000, 411-8; Berglund 1995, 149-59)

Research Value to Society

- Relevant goals, appropriate methods (Tai and Lin 2001, 51-4)
- Risks and benefits (Wu 1994, 367-71)
- Social, political, and economic milieu (Bian and Keller 1999, 439-52)

Clinical Research Practices

- Clinical trials (Chassany and Duracinsky 1999, 437-44; DeMets 2002, 2779-87; Greco 1999, 33-4; Slack et al. 2000, 291-5; Sweatman 2003, 1-5; Taylor 1979, 981-3)
- The use of placebos in clinical trials (Carpenter et al. 2003, 356-62; Vierira 2002, 264; Lau and Woo 2003, 192-8)
- Research practices in the developed countries compared with those in developing countries (Sweatman 2003, 1-5)
- The Declaration of Helsinki, its current version and interpretation (Zion et al. 2000, 615-7; Rothman 2000, 442-5; Greco 2000, 98-101; Botbol-Baum 2000, 238-45)

The ultimate goals are to conduct ethical research and to minimize the risks to human subjects whatever the setting. The challenge facing investigators is to design studies within an ethical framework that is legally acceptable to both the host country and the U.S. sponsors. Recent failures to adequately protect subjects have prompted some countries to amend their policies making compliance more challenging both for local and international scientists.

Case Studies

The ethical questions surrounding conducting research internationally have been vigorously discussed. These case studies illustrate just a few of the issues that have been raised.

AZT placebo trials

In the early 1990s in trials done in the United States and France researchers established that treating HIV-positive women with antiretroviral therapy (Zidovudine (AZT) ACTG 076 trial) decreases HIV transmission from mother to child during birth by two thirds (Lange 1994, 10-1). The ACTG 076 regimen, which involves prenatal and perinatal treatment of the mother and administration of AZT to the newborn, quickly became the standard of care for HIV-infected women and their newborns in the US and other more developed nations.

There are more women infected with HIV in Africa than any other continent in the world (an estimated 15 million at the end of 2001). The life-saving potential of AZT in Africa and in other developing countries with high prevalence of HIV is tremendous. However, the cost of a single treatment far exceeds the average per capita expenditure on health care in most developing countries. In an effort to develop an affordable and effective treatment that would compare in efficacy to the AZT ACTG 076 regimen, investigators from 11 developing nations, in collaboration with UNAIDS, WHO, Centers for Disease Control (CDC) and National Institutes of Health (NIH) researchers, conducted studies in Africa and Thailand. These studies were designed to compare a shorter, less intensive and therefore less expensive regime of AZT to a placebo control. A placebo control was considered acceptable since the standard of care for women with HIV in these countries was no treatment at all. These studies reported that, compared to placebo, the short course of AZT reduced the maternal transmission of HIV by 50% in the absence of breastfeeding (Shaffer et al. 1999, 773-80) and 38% when mothers breastfed their newborns (Dabisa et al. 1999, 786-92).

These trials generated considerable debate about whether the use of a placebo in this research design was ethically acceptable considering there was a course of effective treatment used elsewhere in the world, but unavailable in the countries in which the short course was conducted (Bhagat and Nyazema 2000, 105-7; Lurie and Wolfe 1997, 853-6).

Ethical concerns: The use of placebo study group, when a treatment does exist somewhere, and the inclusion of a vulnerable population.

Surfaxin study

Another recent clinical study that has been debated is the trial examining the drug Surfaxin to treat infants with respiratory distress syndrome. As in the AZT study, a proven effective treatment already existed in the United States. Therefore any study examining an intervention for respiratory distress syndrome with a placebo control would be ethically unacceptable in the U.S. as it would have required withholding proven effective treatment from some infants. The company proposed conducting this study in Latin America where infants with lung illnesses do not usually have access to effective treatment (Charatan 2001, 575).

Although this study is similar in many ways to the AZT study, there are a few significant differences. One of those differences concerns the research participants themselves. Whereas the women participating in the AZT study were presumably competent adults with the ability to give voluntary consent for themselves and their infants, in the Surfaxin study the infant subjects could not consent for themselves and would require added protections according to the principle of respect for persons.

Another important difference is the distribution of benefits and burdens inherent in the research design. It was always the intent of the manufacturer to market Surfaxin in the U.S. Thus, the benefits would have gone to infants in the U.S. and perhaps other more developed countries while the burdens (risks) of the research would be borne by infants in Latin America. This distinction between the population asked to test the drug and the population in which the drug would be marketed violates the principle of distributive justice (i.e., an unequal distribution of benefit and burden).

Redesigning the study might have improved its ethical foundation. For example, the drug might have been provided to the Latin American country free for a 10-year period, or at a discounted price. For those infants and families in Latin America with no access to certain types of health care, these types of studies may offer Latin American families their only access to a potentially lifesaving option (Jesus and Higgs 2002, 93-5).

Ethical concerns: Vulnerable population, unequal distribution of benefit and burden.

Anhui Studies

In 1999, a former faculty member of the Harvard School of Public Health (HSPH) complained to the OHRP that subjects in several collaborative observational epidemiological projects conducted in the remote rural area of Anhui Province, China, by HSPH, the Brigham and Women's Hospital, Massachusetts Mental Health Institute, and the Anhui Medical University were being exploited. The researchers, who included Chinese researchers employed by HSPH, had identified this area of China as one with very few genetic variations and considered it an important resource for the study of gene-environment of diseases such as asthma, lung disease, hypertension, and adverse pregnancy outcomes. The Chinese collaborators and Chinese authorities agreed that this area was appropriate for study. The subjects were asked for consent to participate.

Questions raised by the complainant included whether the participants, some of whom were illiterate, truly consented or whether the amount they were paid could be considered an inappropriate inducement. In 2003, the OHRP found that the IRB at the Harvard School of Public Health (HSPH) had not documented its ethical review of the studies properly. As a result of the investigation, HSPH put into effect new policies concerning training and a quality improvement program for its human subject protections system.

Lessons from this experience are not specific to China, but have to do with mounting a robust, well-staffed effort to meet the evolving regulatory needs around human subjects protection. Domestic and foreign researchers should consider whether some people, especially in rural areas, are capable of consenting to certain research projects.

Ethical concerns Lack of proper informed consent and the involvement of a vulnerable population.

Study Review Guidelines

U.S. Government Guidelines Concerning Ethical Review



- 45 CFR 46, Subpart A, (the "Common Rule") requires that all human research funded by the federal government be reviewed by an Institutional Review Board (IRB) or board of ethics.
- When federal funds are awarded to a U.S. institution to conduct research outside the U.S., in most cases the protocol will need to be reviewed by an IRB in both the U.S. and the host country.
- Research conducted by private companies on test articles that will subsequently be submitted through the U.S. Food and Drug Administration (FDA) must be reviewed by an IRB. However, U.S. regulations do not specify which country's IRB should conduct the review.

Ethical Review in Host Country

For studies subject to 45 CFR 46, U.S. regulations require that a study must be reviewed and approved by an IRB in the area where the study will be conducted. It is not enough that a U.S.-based IRB approve a study; it must also be approved by a board made up of members of the community in which the study will be performed. This requirement is in addition to complying with host country national guidelines or regulations.

Host-country national approval

National systems for the approval of human subjects research vary widely from country to country. Some countries (for example, India and The Gambia) have clear and robust guidelines for externally sponsored research as well as for domestic

projects while other countries do not have review committees responsible for human subjects protection. In such cases the country may have other means of approving human subjects research such as review by the Ministry of Health (MOH). For additional information about host country requirements refer to OHRP's International Compilation of Human Research Standards.

Countries without specific guidelines

Countries that do not have review boards may request assistance from neighboring countries. For example, in Guinea Bissau, once the government has considered a research proposal and decided that it is appropriate, it is sent to the National Ethics Committee in The Gambia for review and approval. Some Latin American countries have regulatory procedures that require both local scientific approval and approval by an independent ethics committee. Countries in Central and Eastern Europe also vary widely with the respect to IRB or Ethics Board review, from having established national committees to review foreign supported research to being in the process of developing and implementing national policies, procedures, and national IRBs.

Each investigator has the responsibility to clearly understand the mechanism and process of review of human subjects research in the host country.

Local IRBs

U.S. policy is that the local IRB reviewing a research study can be from the collaborating institution, another institution in the same geographical area or by an independent IRB.

Local IRB Requirements

- Register with OHRP. See step-by-step instruction for registering an IRB.
- Assure OHRP that the local IRB will follow the current guidelines including the review of projects.

Forming/Recognizing Existing IRBs

- When an ethical review board exists at the host institution and meets the U.S. criteria for an IRB it can be registered and used as the IRB of record for the study.
- A local independent IRB can also be registered and used as the IRB of record for the study.
- If an ethical review board does not exist, the U.S. collaborator will need to supply information to the foreign collaborator on the makeup and duties of an IRB and assist in the formation of a suitable IRB.

Criteria for IRB Members (Protection of Human Subjects 2017)

- The makeup of the IRB must follow the guidelines used by the U.S.
- The IRB must have a minimum of 5 members including at least one:
  - Scientist from the institution
  - Other person from the institution who is not a scientist
  - A scientist from another organization related to the research (for example, doctor or other researcher)
  - Member of the community at large (for example, teacher or worker)
- The IRB should have a chair person as well as a person designated as the human subjects administrator who is not the IRB chairperson.

Summary

The core ethical concepts embodied in the Belmont Report provide a framework for conducting human subjects research. Interpreting and applying them in another cultural setting may be challenging. International guidelines for international research ethics outline and interpret changes, advances and controversies in research ethics outside western countries. Researchers need a clear understanding of both the ethical guidelines and government regulations pertaining to any proposed research, from both the U.S. and host country governments.

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Additional Resources

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This module has a quiz.

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